Content-based RS – State of the Art and Trends

CB-RS

- Recommender systems have the effect of guiding users in a personalized way to interesting objects in a large space of possible options.
- Content-based recommendation systems try to recommend items similar to those a given user has liked in the past

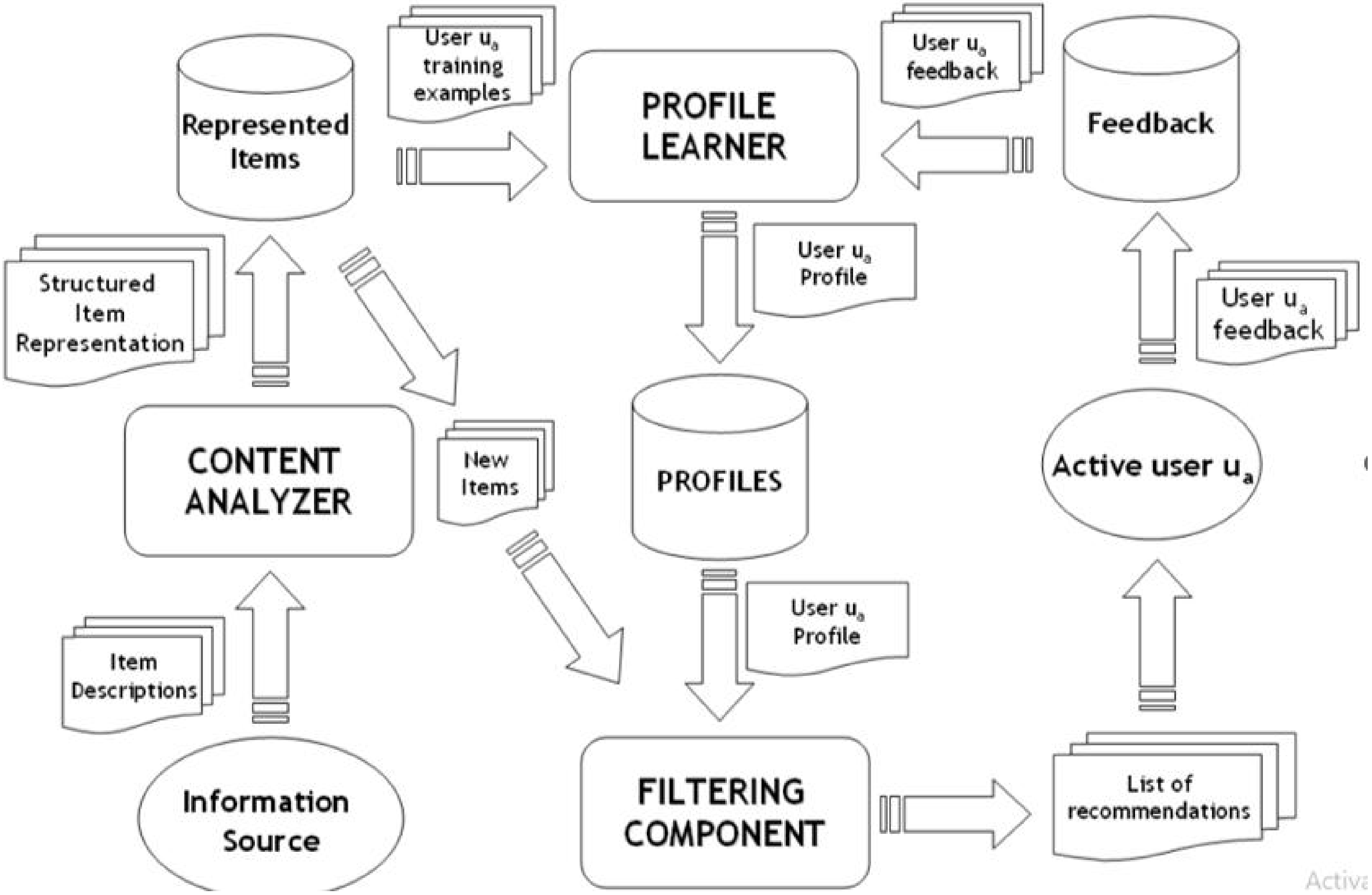

Fig. High level architecture of a Content-based Recommender

Item Representation

- Items that can be recommended to the user are represented by a set of features also called attributes or properties.

- When each item is described by the same set of attributes, and there is a known set of values the attributes may take, the item is represented by means of structured data. In this case, many **ML algorithms** can be used to learn a user profile.

Item Representation..

- In most content-based filtering systems, item descriptions are textual features extracted from **Web pages, emails, news articles or product descriptions**.

- Unlike structured data, there are no attributes with well-defined values.

- Textual features create a **number of complications** when learning a user profile, due to the natural language ambiguity.

- The **problem** is that **traditional keyword-based profiles** are unable to capture the semantics of user interests because they are primarily driven by a string matching operation.

- If a string, or some **morphological variant**, is found in both the profile and the document, a match is made, and the document is considered as relevant.

Item Representation..

- **Semantic analysis** and its integration in personalization models is one of the most innovative and interesting approaches proposed in literature to solve those problems.

- The key idea is the adoption of knowledge bases, such as **lexicons or ontologies** , for annotating items and representing profiles in order to obtain a “semantic” interpretation of the user information needs.

Item Representation..

- **Semantic analysis** and its integration in perso innovative and interesting approaches pro problems.

These formalize concepts, their properties, and interrelations—often used with description logics. Ontologies enable reasoning about "is-a" hierarchies, associations, and domain-specific knowledge. They provide deeper context beyond mere synonyms.

- The key idea is the adoption of knowledge bases, such as **lexicons or ontologies** , for annotating items and representing profiles in order to obtain a "semantic" interpretation of the user information needs.

Semantic Lexicons (e.g., WordNet, EuroWordNet): These are structured dictionaries where words are grouped into synsets (synonym sets) and linked via semantic relations like hypernymy/hyponymy. They help in understanding term meaning, aiding tasks like query expansion and word sense disambiguation.

Keyword-based Vector Space Model

- Most content-based recommender systems use relatively simple retrieval models, such as **keyword matching** or the **Vector Space Model (VSM)** with basic **TF-IDF weighting**.

- VSM is a spatial representation of text documents.

- In that model, each document is represented by a vector in a **n-dimensional space**, where each dimension corresponds to a term from the overall vocabulary of a given document collection.

Keyword-based Vector Space Model

- Formally, every document is represented as a vector of term weights, where each weight indicates the degree of association between the document and the term.

- Let ***D* = {*d*1,*d*2, ...,*dN*}** denote a set of documents or corpus, and ***T* = {*t*1, *t*2, ..., *tn*}** be the dictionary, that is to say the set of words in the corpus.
- **T** is obtained by applying some standard natural language processing operations, such as tokenization, stopwords removal, and stemming.
- Each document d_j is represented as a vector in a n-dimensional vector space, so dj = {w1 j,w2 j, ...,dn j}, where $w_{k\,j}$ is the weight for term t_k in document d_j.

Keyword-based Vector Space Model

- Document representation in the VSM raises two issues: weighting the terms and measuring the feature vector similarity.
- The most commonly used term weighting scheme, **TF-IDF (Term Frequency-Inverse Document Frequency) weighting**, is based on empirical observations regarding text.

- rare terms are not less relevant than frequent terms (IDF assumption);
- multiple occurrences of a term in a document are not less relevant than single occurrences (TF assumption);
- long documents are not preferred to short documents (normalization assumption).

Keyword-based Vector Space Model

- In other words, terms that occur frequently in one document (TF =term-frequency), but rarely in the rest of the corpus (IDF = inverse-document-frequency), are more likely to be relevant to the topic of the document.

- In addition, normalizing the resulting weight vectors prevent longer documents from having a better chance of retrieval.

- These assumptions are well exemplified by the TF-IDF function

$$\text{TF-IDF}(t_k, d_j) = \underbrace{\text{TF}(t_k, d_j)}_{\text{TF}} \cdot \underbrace{log \frac{N}{n_k}}_{\text{IDF}}$$

where N denotes the number of documents in the corpus, and n_k denotes the number of documents in the collection in which the term t_k occurs at least once.

Keyword-based Vector Space Model

$$\mathrm{TF}(t_k, d_j) = \frac{f_{k,j}}{max_z f_{z,j}}$$

- where the maximum is computed over the frequencies *fz, j* of all terms *tz* that occur in document *d*.
- In order for the weights to fall in the [0,1] interval and for the documents to be represented by vectors of equal length, weights obtained by Equation are usually normalized by cosine normal.

$$w_{k,j} = \frac{\text{TF-IDF}(t_k, d_j)}{\sqrt{\sum_{s=1}^{|T|} \text{TF-IDF}(t_s, d_j)^2}}$$

which enforces the normalization assumption.

Keyword-based Vector Space Model

- As stated earlier, a similarity measure is required to determine the closeness between two documents. Many similarity measures have been derived to describe the proximity of two vectors; among those measures, cosine similarity is the most widely used.

$$sim(d_i, d_j) = \frac{\sum_k w_{ki} \cdot w_{kj}}{\sqrt{\sum_k w_{ki}^2} \cdot \sqrt{\sum_k w_{kj}^2}}$$

In content-based recommender systems relying on VSM, both user profiles and items are represented as weighted term vectors. Predictions of a user's interest in a particular item can be derived by computing the cosine similarity

Review of Keyword-based Systems

- Several keyword-based recommender systems have been developed in a relatively short time, and it is possible to find them in various fields of applications, **such as news, music, e-commerce, movies, etc.**
- **In the area of Web recommenders:** famous systems in literature are

Letizia , Personal WebWatcher , Syskill & Webert , ifWeb , Amalthea , and WebMate .

Letizia is implemented as a web-browser extension that tracks the user’s browsing behavior and builds a personalized model consisting of keywords related to the user’s interests.

It relies on implicit feedback to infer the user’s preferences. For example, bookmarking a page is interpreted as strong evidence for the user’s interests in that page.

Review of Keyword-based Systems

- **Personal WebWatcher** learns individual interests of users from the Web pages they visit, and from documents lying one link away from the visited pages.

- It processes visited documents as positive examples of the user's interests, and non-visited documents as negative examples.

- **Amalthaea** uses specific filtering agents to assist users in finding interesting information as well.
- User can specify filtering agents by providing pages (represented as weighted vectors) closely related to their interests

Review of Keyword-based Systems

- In the field of **news filtering**, noteworthy recommender systems are NewT , PSUN , INFOrmer , NewsDude , Daily Learner , and YourNews.

- **NewT (News Tailor)** allows users to provide positive and negative feedback on articles, part of articles, authors or sources.

- Several filtering agents are trained for different types of information, e.g. one for political news, one for sports, etc.

Review of Keyword-based Systems

- Learning short-term and long-term profiles is quite typical of news filtering systems.

- NewsDude learns a short-term user model based on TF-IDF (cosine similarity), and a long-term model based on a naıve Bayesian classifier by relying on an initial training set of interesting news articles provided by the user.

- Explore LIBRA and INTIMMATE RS.

Semantic Analysis by using Ontologies

- Semantic analysis allows learning more accurate profiles that contain references to concepts defined in external knowledge bases.
- The main motivation for this approach is the challenge of providing a recommender system with the cultural and linguistic background knowledge which characterizes the ability of interpreting natural language documents and reasoning on their content.
- The description of these strategies is carried out by taking into account several criteria:
- **the type of knowledge source involved (e.g. lexicon, ontology, etc.);**
- **the techniques adopted for the annotation or representation of the items;**
- **the type of content included in the user profile;**
- **the item-profile matching strategy.**

Semantic Analysis by using Ontologies

- **SiteIF** is a personal agent for a multilingual newsWeb site. Iit was the first system to adopt a sense-based document representation in order to build a model of the user interests.
- **ITR (ITem Recommender)** is a system capable of providing recommendations for items in several domains (e.g., movies, music, books), provided that descriptions of items are available as text documents (e.g. plot summaries, reviews, short abstracts).
- **SEWeP (Semantic Enhancement for Web Personalization)** is a Web personalization system that makes use of both the usage logs and the semantics of a Web site's content in order to personalize it.